CUSTOMER BEHAVIOR ANALYSIS

-Aravind Swamy

1. Project Overview

This project analyzes retail customer shopping behavior using a dataset of 3,900 customer records. The goal is to uncover insights into revenue patterns, purchasing habits, customer segmentation, and the impact of discounts and subscriptions — ultimately supporting data-driven business decisions.

Tools & Technologies Used:

- Python (Pandas) — Data cleaning & feature engineering
- PostgreSQL (via SQLAlchemy + psycopg2) — Data storage & SQL analysis
- SQL (PostgreSQL) — Business question queries
- Power BI — Interactive dashboard visualization

Deliverables

1. **Data Preparation & Modeling (Python):** Clean and transform the raw dataset for analysis.
2. **Data Analysis (SQL):** Organize the data into a structured format, simulate business transactions, and run queries to extract insights on customer segments, loyalty, and purchase drivers.
3. **Visualization & Insights (Power BI):** Build an interactive dashboard that highlights key patterns and trends, enabling stakeholders to make data-driven decisions.
4. **Report and Presentation:** Write a clear project report summarizing your key findings and business recommendations. Prepare a presentation that visually communicates insights and actionable recommendations to stakeholders.
5. **GitHub Repository:** Include all Python scripts, SQL queries, and dashboard files in a well-structured repository.

2. Dataset Description

Source File: customer_shopping_behavior.csv **Total Records:** 3,900 rows | **Columns:** 18 original, 19 after feature engineering

The dataset contains the following columns:

- **customer_id** — Unique customer identifier (int)
- **age** — Customer age ranging from 18 to 70 (int)
- **gender** — Male or Female (string)

- **item_purchased** — Name of the product bought; 25 unique items (string)
- **category** — Product category: Clothing, Accessories, Footwear, or Outerwear (string)
- **purchase_amount** — Transaction value in USD, ranging from $20 to $100 (int)
- **location** — US state; covers all 50 states (string)
- **size** — S, M, L, or XL (string)
- **color** — Product color; 25 unique values (string)
- **season** — Spring, Summer, Fall, or Winter (string)
- **review_rating** — Customer rating from 2.5 to 5.0 (float)
- **subscription_status** — Yes or No (string)
- **shipping_type** — One of 6 shipping options (string)
- **discount_applied** — Yes or No (string)
- **previous_purchases** — Number of prior purchases, from 1 to 50 (int)
- **payment_method** — One of 6 payment methods (string)
- **frequency_of_purchases** — One of 7 frequency categories (string)
- **age_group** *(engineered)* — Young Adult, Adult, Middle-aged, or Senior (string)
- **purchase_frequency_days** *(engineered)* — Purchase cycle converted to number of days (int)

3. Data Cleaning & Preparation (Python / Pandas)

3.1 Missing Values

review_rating had **37 null values** out of 3,900 records. These were imputed using the **median rating per category** to preserve distribution integrity:

```
df['Review Rating'] = df.groupby('Category')['Review Rating']
                .transform(lambda x: x.fillna(x.median()))
```

3.2 Column Standardization

Column names were converted to lowercase with underscores for SQL compatibility, and Purchase Amount (USD) was renamed to purchase_amount.

3.3 Duplicate Column Removal

discount_applied and promo_code_used were found to be **100% identical** across all rows. promo_code_used was dropped to eliminate redundancy.

3.4 Feature Engineering

Age Group — customers were segmented into four equal-sized quartile groups using pd.qcut: Young Adult, Adult, Middle-aged, and Senior.

Purchase Frequency Days — the categorical frequency_of_purchases column was mapped to numeric day values: Weekly = 7, Fortnightly/Bi-Weekly = 14, Monthly = 30, Quarterly/Every 3 Months = 90, and Annually = 365.

3.5 Data Export to PostgreSQL

The cleaned DataFrame was loaded into a PostgreSQL database (customer_behavior) using SQLAlchemy, into a table named customer.

4. SQL Analysis

Ten business questions were answered using SQL queries in PostgreSQL.

Q1 — Revenue by Gender: Aggregated total revenue split by Male and Female customers using GROUP BY gender.

Q2 — Discount Users Above Average Spend: Identified customers who used a discount yet still spent at or above the dataset average, using a subquery for the average.

Q3 — Top 5 Products by Review Rating: Ranked all 25 products by average review rating (rounded to 2 decimal places), returning the top 5.

Q4 — Shipping Type vs. Average Purchase Amount: Compared average spend between Standard and Express shipping customers.

Q5 — Subscriber vs. Non-Subscriber Spend: Compared total customers, average spend, and total revenue between subscribed and non-subscribed customers.

Q6 — Products with Highest Discount Rate: Calculated the percentage of transactions with a discount applied per product, returning the top 5 most discounted products.

Q7 — Customer Segmentation: Used a CTE and CASE statement to classify customers as New (1 previous purchase), Returning (2–10), or Loyal (11+).

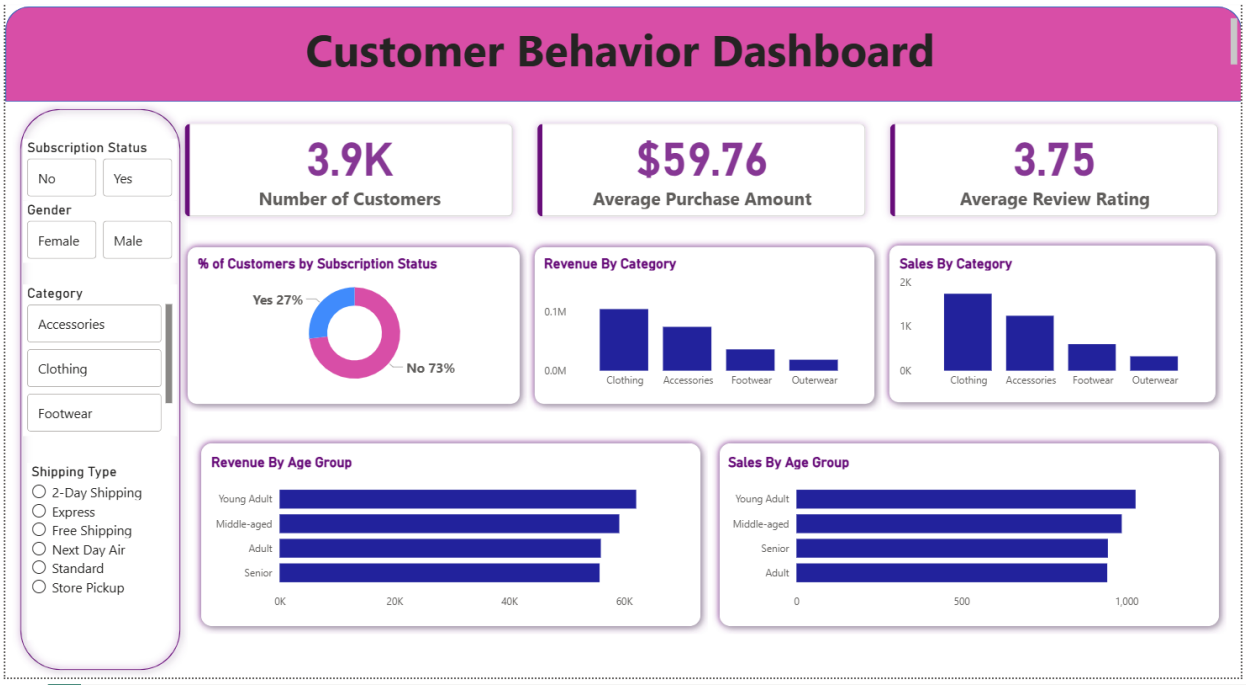
Q8 — Top 3 Products per Category: Used ROW_NUMBER() OVER (PARTITION BY category) to rank products by order count within each of the 4 categories.

Q9 — Repeat Buyers and Subscription Likelihood: Filtered customers with more than 5 previous purchases and grouped by subscription status to assess correlation.

Q10 — Revenue by Age Group: Summed total revenue per age group, ordered by highest revenue.

5. Power BI Dashboard

An interactive dashboard titled **"Customer Behavior Dashboard"** was built in Power BI.



KPI Cards

- **Number of Customers:** 3.9K
- **Average Purchase Amount:** $59.76
- **Average Review Rating:** 3.75

Visualizations

- **% of Customers by Subscription Status** — Donut chart showing 73% non-subscribed vs. 27% subscribed
- **Revenue by Category** — Bar chart; Clothing leads, followed by Accessories, Footwear, and Outerwear
- **Sales by Category** — Bar chart showing transaction count per category
- **Revenue by Age Group** — Horizontal bar chart; Young Adults generate the most revenue
- **Sales by Age Group** — Horizontal bar chart; Young Adults also lead in transaction count

Filters / Slicers

Users can interactively filter all visuals by Subscription Status, Gender, Category, and Shipping Type.

6. Key Findings

- **Clothing** is the top revenue-generating and most-purchased category.
- **Young Adults** lead in both revenue contribution and number of purchases across all age groups.
- Only **27% of customers** are subscribers, representing a significant growth opportunity.
- Customers who applied discounts were still capable of spending above average, suggesting discounts do not cannibalize higher-value purchases.
- The discount_applied and promo_code_used columns were entirely redundant — a data quality finding worth flagging to upstream teams.